Year 12 Further Maths - Circular motion

Similar questions - Strings, tensions and conical motion

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 One end of a light inextensible string of length 1.4 m is attached to a fixed point O on a smooth horizontal table. The other end is attached to a particle P of mass 0.6 kg which moves, with the string taut, in a circular path around O. The speed of P is constant and P completes each circle in 0.7 seconds.

(a) Find the angular velocity of P. **[2]**

(b) Find the speed of P. **[1]**

(c) Find the magnitude of the acceleration of P, and state its direction. **[3]**

(d) The string will break if the tension exceeds 150 N. Find the minimum possible time in which P can complete a circle. **[4]**

2 Two light inextensible strings S_1 and S_2 are each of length 3 m. Two identical particles P and Q each have mass 2 kg. One end of S_1 is attached to P and the other end is attached to a fixed point A on a smooth horizontal table. P moves with constant speed in a horizontal circle with centre A. One end of S_2 is attached to Q and the other end to a fixed point B. Q moves with constant speed in a horizontal circle with centre vertically below B; at any instant BQ makes an angle θ with the downward vertical.

(a) Given that the angular speed of P is the same as the angular speed of Q, show that the tensions in S_1 and S_2 have the same magnitude. **[3]**

(b) You are given instead that the kinetic energy of P is 15 J and that this is the same as the kinetic energy of Q. Find the time for P to complete one revolution and the time for Q to complete one revolution. **[8]**

3 A particle P of mass 0.5 kg is attached to one end of a light inextensible string of length 2.5 m. The other end is attached to a fixed point O on a smooth horizontal plane. P moves in a circle with O as centre.

(a) Find the tension in the string when P moves with angular speed 2 $\mathrm{rad\,s^{-1}}$. **[2]**

A force is now applied to P so that its angular speed increases. When the angular speed reaches 6 $\mathrm{rad\,s^{-1}}$ the string breaks.

(b) Find the speed of P and the tension in the string at the instant the string breaks. **[3]**

After the string breaks, P moves directly up a smooth slope fixed to the plane and inclined at θ° above the horizontal. P travels 20 m up the slope before coming to instantaneous rest.

(c) Use an energy method to find θ. **[3]**

4 Two particles A and B are connected by a light inextensible string. Particle A has mass 0.9 kg and moves on a smooth horizontal table in a circular path of radius 0.5 m and centre O. The string passes through a small smooth hole at O. Particle B moves in a horizontal circle below the table, always vertically below the level of A's circle, with the same angular speed as A. The part of the string below the table has length 1.2 m and makes an angle θ with the downward vertical through O, where $\cos\theta = 0.6$.

(a) Find the time taken for the particles to perform a complete revolution. **[7]**

(b) Find the mass of B. **[3]**

5 AB is a long thin smooth rod fixed vertically with A above B. A light inextensible string of length 1.3 m has one end attached to A and the other end to a particle P of mass m_1 kg. A second string of length 1.3 m has one end attached to P and the other end to a small smooth ring R, of mass m_2 kg, which is free to move on AB. P moves in a horizontal circle of radius 0.5 m with constant angular velocity ω $\mathrm{rad\,s^{-1}}$, with R below P and in equilibrium. The tension in AP is T_1 N and the tension in PR is T_2 N.

(a) By considering the forces on R, express T_2 in terms of m_2. **[2]**

(b) Show that

(i) $T_1 = \dfrac{637}{60}(m_1 + m_2)$, **[2]**

(ii) $\omega^2 = \dfrac{49(m_1 + 2m_2)}{6m_1}$. **[3]**

(c) Deduce that when m_1 is much bigger than m_2, $\omega \approx 2.9$. **[2]**

(d) In a different case $m_1 = 3$ and $m_2 = 2$. The system slows down and eventually comes to rest with P and R hanging in equilibrium vertically below A. Find the total energy lost by P and R as the angular speed of P changes from its initial value to zero. **[5]**

Question 1

(a) $\omega = \dfrac{2\pi}{0.7} = 8.98 \text{ rad s}^{-1}$

(b) $v = r\omega = 1.4(8.976) = 12.6 \text{ m s}^{-1}$

(c) $a = r\omega^2 = 1.4(8.976)^2 = 113 \text{ m s}^{-2}$, directed towards O

(d) $T = mr\omega^2 \leqslant 150$: $\omega^2 \leqslant \dfrac{150}{0.6(1.4)} = 178.6$, $\omega \leqslant 13.36$. Minimum time $= \dfrac{2\pi}{13.36} = 0.470$ s

Question 2

(a) S_1: the tension is the only horizontal force on P, so $T_1 = m\omega^2(3)$. S_2: radius of Q's circle is $3\sin\theta$ and horizontally $T_2\sin\theta = m\omega^2(3\sin\theta)$, so $T_2 = 3m\omega^2 = T_1$

(b) P: $\frac{1}{2}(2)v^2 = 15$, so $v^2 = 15$ and $v = 3.873$; period $= \dfrac{2\pi(3)}{3.873} = 4.87$ s. Q has the same kinetic energy and so the same speed. $T\cos\theta = 2g$ and $T\sin\theta = \dfrac{2v^2}{3\sin\theta}$, so $v^2 = 3g\sin\theta\tan\theta$. Then $15 = 29.4\dfrac{1-\cos^2\theta}{\cos\theta}$, i.e. $\cos^2\theta + \frac{25}{49}\cos\theta - 1 = 0$ and $\cos\theta = \dfrac{-25+\sqrt{10229}}{98} = 0.777$, $\sin\theta = 0.630$. Radius $= 3\sin\theta = 1.889$; $\omega = 3.873/1.889 = 2.051$; period $= \dfrac{2\pi}{\omega} = 3.06$ s

Question 3

(a) $T = mr\omega^2 = 0.5(2.5)(4) = 5$ N

(b) $v = r\omega = 15 \text{ m s}^{-1}$; $T = 0.5(2.5)(36) = 45$ N

(c) $\frac{1}{2}(0.5)(225) = 0.5g(20)\sin\theta$, so $\sin\theta = \dfrac{225}{392} = 0.574$ and $\theta = 35.0$

Question 4

(a) Let the tension be T. A: $T = 0.9\omega^2(0.5)$. B: $T\cos\theta = m_Bg$ and $T\sin\theta = m_B\omega^2(1.2\sin\theta)$, so $T = 1.2m_B\omega^2$. Dividing the vertical equation: $1.2m_B\omega^2(0.6) = m_Bg$, so $\omega^2 = \dfrac{9.8}{0.72} = 13.61$, $\omega = 3.689$. Period $= \dfrac{2\pi}{\omega} = 1.70$ s

(b) $T = 0.9(13.61)(0.5) = 6.125$ N; $m_B = \dfrac{T\cos\theta}{g} = \dfrac{6.125(0.6)}{9.8} = 0.375$ kg

Question 5

(a) The ring is smooth, so the rod exerts only a horizontal force. Vertically, with $\cos\alpha = \dfrac{1.2}{1.3} = \dfrac{12}{13}$ the angle of PR to the vertical: $T_2\cos\alpha = m_2g$, so $T_2 = \dfrac{13}{12}m_2g = \dfrac{637}{60}m_2$

(b)(i) For P vertically: $T_1\cos\alpha = m_1g + T_2\cos\alpha = (m_1+m_2)g$, so $T_1 = \dfrac{13g}{12}(m_1+m_2) = \dfrac{637}{60}(m_1+m_2)$

(ii) Horizontally: $(T_1+T_2)\sin\alpha = m_1\omega^2(0.5)$ with $\sin\alpha = \frac{5}{13}$. $T_1+T_2 = \dfrac{13g}{12}(m_1+2m_2)$, so $\dfrac{5g}{12}(m_1+2m_2) = 0.5m_1\omega^2$ and $\omega^2 = \dfrac{5g(m_1+2m_2)}{6m_1} = \dfrac{49(m_1+2m_2)}{6m_1}$

(c) $m_2/m_1 \to 0$: $\omega^2 \approx \dfrac{49}{6} = 8.17$, so $\omega \approx 2.86 \approx 2.9$

(d) Initially $\omega^2 = \dfrac{49(7)}{18} = 19.06$, so $v^2 = \omega^2(0.25) = 4.764$ and KE $= \frac{1}{2}(3)(4.764) = 7.146$ J. Initially P is 1.2 m and R is 2.4 m below A. Finally P is 1.3 m and R is 2.6 m below A, so P and R descend 0.1 m and 0.2 m. PE lost $= 3g(0.1) + 2g(0.2) = 6.86$ J. Energy lost $= 7.146 + 6.86 = 14.0$ J